# THE INDIAN INTERNATIONAL SCHOOL,DSO
## Worksheet – Chapter 4: Determinants (2026-2027)

Grade: 12 Subject: MATHEMATICS

Name: _____

Date: _____

**Instructions:**

- This worksheet has 20 questions arranged in three levels of increasing difficulty.
- Level 1: Q 1 – Q 7 (MCQs, Assertion–Reason and short answer questions)
- Level 2: Q 8 – Q 14 (short answer questions)
- Level 3: Q 15 – Q 20 (long answer and case-based questions)
- Marks for each question are given against it. All questions are compulsory.
- The answer key is given at the end of the worksheet.

| Q No. | Question | Marks |
|---|---|---|
| | **LEVEL 1**<br>**Q 1 – Q 7** | **8** |
| 1. | If $\begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix}$, then $x$ is equal to:<br>(A) 6 (B) $\pm 6$ (C) $-6$ (D) 0 | 1 |
| 2. | If $A$ is a square matrix of order 3 with $\lvert A\rvert = 5$, then $\lvert \text{adj } A\rvert$ is:<br>(A) 5 (B) 25 (C) 125 (D) $\frac{1}{5}$ | 1 |
| 3. | If $A$ is a square matrix of order 3 and $\lvert A\rvert = 4$, then $\lvert 2A\rvert$ is:<br>(A) 8 (B) 16 (C) 32 (D) 64 | 1 |
| 4. | The area of the triangle with vertices $(-2,-3)$, $(3,2)$ and $(-1,-8)$ is:<br>(A) 15 sq units (B) 30 sq units<br>(C) 20 sq units (D) 10 sq units | 1 |
| 5. | The cofactor of the element $a_{12}$ in $\begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix}$ is:<br>(A) 46 (B) $-46$ (C) 4 (D) $-4$ | 1 |

| Q No. | Question | Marks |
|---|---|---|
| 6. | **Assertion (A):** If $A$ is a $3 \times 3$ matrix with $\|A\| = 2$, then $\|A\ (\text{adj}\ A)\| = 8$.<br>**Reason (R):** For any square matrix $A$, $A\ (\text{adj}\ A) = \|A\|\, I$.<br>(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).<br>(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).<br>(C) Assertion (A) is true, but Reason (R) is false.<br>(D) Assertion (A) is false, but Reason (R) is true. | 1 |
| 7. | Find the values of $k$ if the area of the triangle with vertices $(k, 0)$, $(4,0)$ and $(0,2)$ is 4 sq units. | 2 |
| | **LEVEL 2**<br>**Q 8 – Q 14** | **17** |
| 8. | Find the adjoint and inverse of $A = \begin{bmatrix} 2 & -1 \\ 4 & 3 \end{bmatrix}$. | 2 |
| 9. | If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, show that $A^2 - 5A + 7I = O$. Hence find $A^{-1}$. | 3 |
| 10. | Using determinants, find the equation of the line joining the points $(1,2)$ and $(3,6)$. | 2 |
| 11. | Solve the system of equations using the matrix method: $5x + 2y = 4$, $7x + 3y = 5$. | 3 |
| 12. | Examine the consistency of the system: $x + 3y = 5$, $2x + 6y = 8$. | 2 |
| 13. | If $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$, show that $A^{-1} = \frac{1}{19}A$. | 2 |
| 14. | If $A = \begin{bmatrix} 2 & 3 \\ 1 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$, verify that $(AB)^{-1} = B^{-1}A^{-1}$. | 3 |
| | **LEVEL 3**<br>**Q 15 – Q 20** | **29** |
| 15. | A school wants to allocate students into three clubs: Sports, Music and Drama, under the following conditions: (i) the number of students in the Sports club should be equal to the sum of the numbers of students in the Music and Drama clubs; (ii) the number of students in the Music club should be 20 more than half the number of students in the Sports club; (iii) the total number of students allocated to the three clubs is 180. Using the matrix method, find the number of students allocated to each club. | 5 |
| 16. | Find the product $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}\begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}$ and use it to solve: $x - y + 2z = 1$, $2y - 3z = 1$, $3x - 2y + 4z = 2$. | 5 |
| 17. | If $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{bmatrix}$, find $A^{-1}$. Hence solve: $x + 2y - 3z = -4$, $2x + 3y + 2z = 2$, $3x - 3y - 4z = 11$. | 5 |
| 18. | The cost of 4 kg onion, 3 kg wheat and 2 kg rice is ₹60. The cost of 2 kg onion, 4 kg wheat and 6 kg rice is ₹90. The cost of 6 kg onion, 2 kg wheat and 3 kg rice is ₹70. Find the cost of each item per kg by the matrix method. | 5 |

| Q No. | Question | Marks |
|---|---|---|
| 19. | If $A = \begin{bmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{bmatrix}$, find $A^{-1}$. Using $A^{-1}$, solve: $x - 2y = 10$, $2x - y - z = 8$, $-2y + z = 7$. | 5 |
| 20. | **Case Study:** A farmer owns a triangular plot of land whose corners, on a map grid, are $A(2,3)$, $B(-1,0)$ and $C(3,-2)$ (1 unit = 10 m). He plans to fix a pole at a point $D(k,1)$ on the boundary line $AB$.<br>**(i)** Using determinants, find the area of the plot in square units. **[1]**<br>**(ii)** Find the value of $k$ so that $D$ lies on line $AB$. **[1]**<br>**(iii)** Using determinants, find the equation of line $AB$. **[2]** | 4 |

***** End of Worksheet *****

# ANSWER KEY

## Chapter 4: Determinants

| Q No. | Answer / Key Steps |
|---|---|
| | **LEVEL 1** |
| 1. | (B) $x^2 - 36 = 0 \Rightarrow x = \pm 6$ |
| 2. | (B) $\lvert\text{adj}A\rvert = \lvert A\rvert^2 = 25$ |
| 3. | (C) $2^3 \times 4 = 32$ |
| 4. | (A) $\frac{1}{2}\lvert -20 - 15 + 5\rvert = 15$ |
| 5. | (A) $A_{12} = -(6(-7) - 4 \cdot 1) = 46$ |
| 6. | (A) Both true; R explains A. |
| 7. | $\frac{1}{2}\lvert 8 - 2k\rvert = 4 \Rightarrow k = 0$ or $k = 8$ |
| | **LEVEL 2** |
| 8. | $\text{adj}A = \begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix}$, $\lvert A\rvert = 10$, $A^{-1} = \frac{1}{10}\begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix}$ |
| 9. | $A^2 = \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix}$; $A^2 - 5A + 7I = O$. $A^{-1} = \frac{1}{7}(5I - A) = \frac{1}{7}\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$ |
| 10. | $\begin{vmatrix} x & y & 1 \\ 1 & 2 & 1 \\ 3 & 6 & 1 \end{vmatrix} = 0 \Rightarrow y = 2x$ |
| 11. | $\lvert A\rvert = 1$; $x = 2$, $y = -3$ |
| 12. | $\lvert A\rvert = 0$ and $(\text{adj}A)B = \begin{bmatrix} 6 \\ -2 \end{bmatrix} \neq O$; inconsistent. |
| 13. | $\lvert A\rvert = -19$, $\text{adj}A = \begin{bmatrix} -2 & -3 \\ -5 & 2 \end{bmatrix}$, $A^{-1} = \frac{1}{19}\begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix} = \frac{1}{19}A$ |
| 14. | $AB = \begin{bmatrix} -1 & 5 \\ 5 & -14 \end{bmatrix}$, $(AB)^{-1} = \frac{1}{11}\begin{bmatrix} 14 & 5 \\ 5 & 1 \end{bmatrix}$; $B^{-1} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$, $A^{-1} = \frac{1}{11}\begin{bmatrix} 4 & 3 \\ 1 & -2 \end{bmatrix}$; $B^{-1}A^{-1} = \frac{1}{11}\begin{bmatrix} 14 & 5 \\ 5 & 1 \end{bmatrix}$. Verified. |
| | **LEVEL 3** |
| 15. | Equations: $x - y - z = 0$, $-x + 2y = 40$, $x + y + z = 180$ ($x$: Sports, $y$: Music, $z$: Drama). $\lvert A\rvert = -4 \neq 0$; $X = A^{-1}B$ gives Sports $= 90$, Music $= 65$, Drama $= 25$. |
| 16. | Product $= I$, so $A^{-1} = \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}$; $X = A^{-1}B$: $x = 0$, $y = 5$, $z = 3$ |
| 17. | $\lvert A\rvert = 67$, $A^{-1} = \frac{1}{67}\begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$; $x = 3$, $y = -2$, $z = 1$ |
| 18. | Onion ₹5/kg, wheat ₹8/kg, rice ₹8/kg |
| 19. | $\lvert A\rvert = 1$, $A^{-1} = \begin{bmatrix} -3 & -2 & -4 \\ 2 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}$. The coefficient matrix is $A'$, so $X = (A^{-1})'B$: $x = 0$, $y = -5$, $z = -3$ |
| 20. | (i) $\frac{1}{2}\lvert 2(0 + 2) - 1(-2 - 3) + 3(3 - 0)\rvert = 9$ sq units. (ii) $\begin{vmatrix} 2 & 3 & 1 \\ -1 & 0 & 1 \\ k & 1 & 1 \end{vmatrix} = 3k = 0 \Rightarrow k = 0$. (iii) $\begin{vmatrix} x & y & 1 \\ 2 & 3 & 1 \\ -1 & 0 & 1 \end{vmatrix} = 0 \Rightarrow x - y + 1 = 0$ |